LEARNER GUIDE

Volume

2D and 3D Shapes. A short guide to walk beside you — read as much or as little as helps.
Connection before curriculum. Always.

What this lesson is about

Volume is the amount of 3D space a solid fills, measured in cubic units (cm^3). In this lesson you build a prism by carrying a cross-section along a length and watch its volume grow as area × length; you fill solids with unit cubes, layer by layer; and you discover that leaning a stack of layers over does not change how much it holds. You will find the volume of cuboids, prisms and cylinders, and meet litres along the way. The question you are exploring is: how much space does a solid fill?

What you are learning to notice

- That volume is the space inside a solid, measured in cubic units (cm^3).
- That the volume of any prism is its cross-section area × length.
- That a cuboid is length × width × height, and a cylinder is πr^2 × height.
- That leaning a stack of layers keeps the same volume — and that 1 litre = 1000 cm^3.

Moving through it, one step at a time

In the Prism Volume Machine, choose a cross-section, read its area, and drag the length to watch volume = area × length. In the Layer-Stacker, set the base and the number of layers to fill a solid with unit cubes, then drag Lean to slide the layers over. In Spot the Slip, decide whether each volume calculation is right or has a mistake. Then work out how much real containers hold, in cm^3 and litres. There is no rush; use the isometric grid on the Scratchpad to sketch solids.

> **A gentle reminder**
> You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the area of a rectangle and a circle (πr^2), cross-sections and prisms (L8), and multiplication), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Learner Guide v0.1.